



Causal Discovery for Vector-valued Variables and Consistency-guided Aggregation

Urmi Ninad¹

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(work with Jonas Wahl^{2*}, Andreas Gerhardus^{3*}, Jakob Runge⁴)

*: equal contribution

1: Technische Universität Berlin

2: German Research Centre for Artificial Intelligence (DFKI)

3: German Aerospace Center (DLR), Institute of Data Science

4: Department of Computer Science, University of Potsdam



- Why vector-valued causal discovery (CD)?
- Vector-valued data-generation processes
- Three approaches to CD with vector-valued variables
- Valid aggregation maps for CD
- Aggregation consistency scores and Adag wrapper
- Conclusions

Why vector-valued CD?

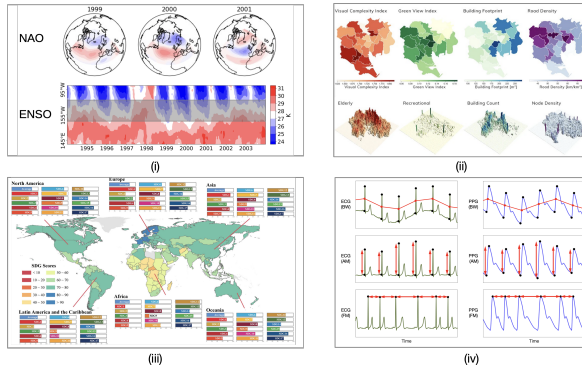


Figure: Examples of scientific domains exhibiting vector-valued data

- (i) Earth science, eg. spatiotemporal systems such as NAO and ENSO,
- (ii) Urban systems, eg. multiple variables represent aspects of cities [Yap and Biljecki, 2023]
- (iii) Economics, eg. multiple indices describe notions of sustainable development [Zhao et al., 2022]
- (iv) Respiratory signals of the human body which are represented by spike interval length and amplitude (image from wikipedia commons).

Assumption (Existence of vector-valued SCM)

Vector-valued variables $\mathbf{X}^i$, jointly independent noises $\boldsymbol{\eta}^i$, and functions $f^i(\cdot)$ such that:

$$\mathbf{X}^i := f^i(Pa(\mathbf{X}^i), \boldsymbol{\eta}^i) . \quad (1)$$

SCM (1) entails graph $\mathcal{G}$ and joint distribution $P(\mathbf{X})$.

- Arguably strong(est) assumption as it requires vector(-valued) variables to have a causal interpretation

¹[Wahl et al., 2024, Anand et al., 2023]

Modeling vector-valued data-generation processes

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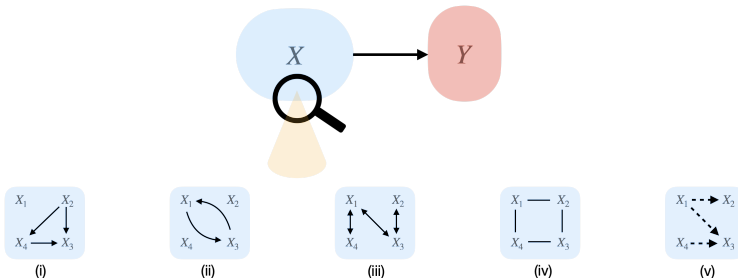
We focus on constraint-based CD so we need further enabling assumptions:

Assumption (CD assumptions)

Graph $\mathcal{G}$ is acyclic and faithful¹ w.r.t. $P(\mathbf{X})$.

¹[Wahl et al., 2024, Anand et al., 2023]

Possibilities for internal dynamics



Given this setting, there might be a few things happening internal to the variables:

- (i) internal DAG
- (ii) internal cycles
- (iii) internal latent confounding
- (iv) internal relationships 'acausal' (eg. equilibrium state of random field)
- (v) internal faithfulness violations, eg. due to determinism
- (vi) combination of the above

These different dynamics invite different approaches to vector-valued CD.

Approaches to CD in vector-valued setting

Common constraint-based approaches to 'vector CD':

1. **Component-wise approach**

- Treats individual components of each vector-valued variables as causal variables
- Performs CD over all components and maps the micro graph to macro graph using user-defined edge-aggregation map
- lower dimensional CI tests but many in number; selectively sound depending on internal dynamics

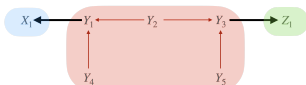
2. **Aggregation approach**

- Considers aggregates of each variable (eg. averages), then performs CD over aggregates
- sample efficient but soundness not guaranteed

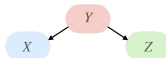
3. **Vectorized approach**

- Runs CD over vector-valued variables directly
- theoretically sound but relies on multivariate CI tests, a harder problem than univariate testing

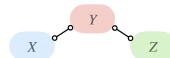
Examples to illustrate approaches



(a) True graph

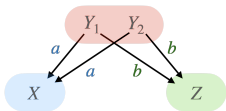


(b) Component-wise result

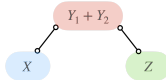


(c) Vectorized result

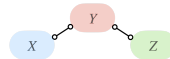
Figure: Component-wise CD is sound when internal dynamics correspond to the CD assumptions; might result in more orientations



(a) True graph



(b) Aggregate result



(c) Vectorized result

Figure: Here aggregate CD is sound because coefficients are fine-tuned.

Experiments to illustrate pitfalls of approaches I

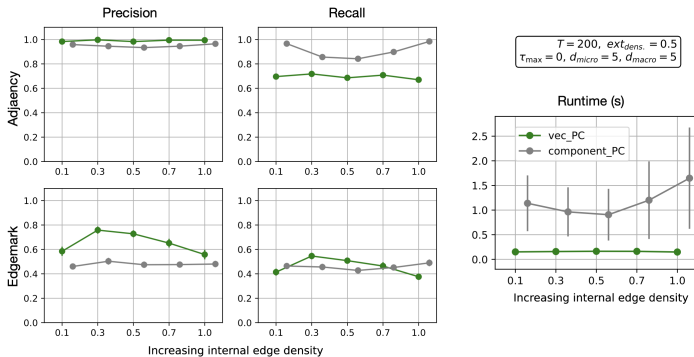
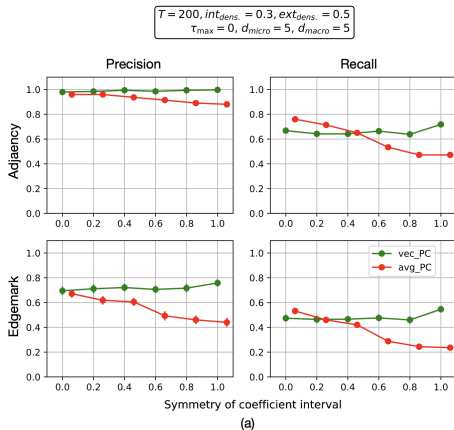


Figure: Comparing vectorized-PC with component-wise PC and majority edge-aggregation strategy. Higher recall but lower precision of component-wise PC implies more edges are found in general

Experiments to illustrate pitfalls of approaches II

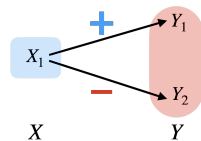
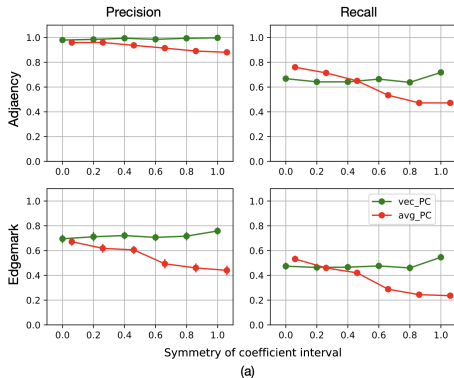


Comparing vectorized-PC with aggregate-PC (aggregation map
≡ average)

Experiments to illustrate pitfalls of approaches II

$$T = 200, \text{int}_{dens.} = 0.3, \text{ext}_{dens.} = 0.5$$

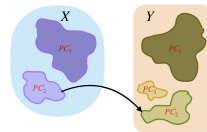
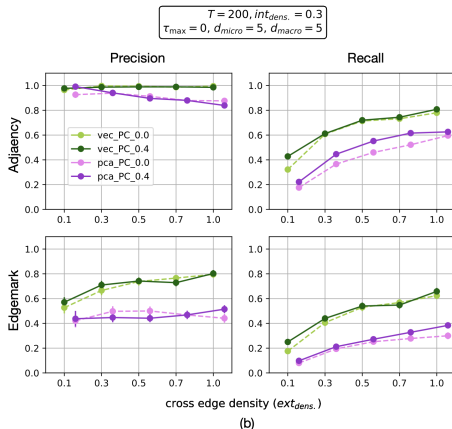
$$\tau_{\max} = 0, d_{\text{micro}} = 5, d_{\text{macro}} = 5$$



As coefficient interval becomes symmetric around zero, more cancellation of effects

Comparing vectorized-PC with aggregate-PC (aggregation map $\equiv$ average)

Experiments to illustrate pitfalls of approaches III



Dominant causal
relationship between
lower principal
components

Comparing vectorized-PC with aggregate-PC (aggregation map
 $\equiv$ 1st principal component of each variable)

Aggregate approach and consistent causal abstractions

- Aggregation maps have been dealt with within the consistent abstraction framework² that aims to establish criteria to map low-level SCM to high-level SCM consistently.
- We argue that if aim is to learn causal graph $\mathcal{G}$, one needn't hold up aggregation map to strict criteria of consistency among SCMs, but the criteria may be loosened.

²[Rubenstein* et al., 2017]

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- Especially important for fields where interventional data is categorically unavailable, eg. earth system.

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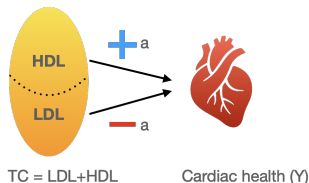


Figure: TC may not a causal variable consistent with the underlying SCM, but a helpful aggregate variable for causal discovery.

²[[Rubenstein* et al., 2017](#)]

Validity of aggregation maps for CD

- Vector-valued variables $\mathbf{X}^i$, together denoted by $\mathbf{X}$
- Aggregation map $g(\cdot)$, such that $g(\mathbf{X}^i) = \mathbf{Z}^i$
- If $\mathcal{S}_{\mathbf{X}} = (\mathbf{X}^{i_1}, \dots, \mathbf{X}^{i_k}) \subseteq \mathbf{X}$, then $g(\mathcal{S}_{\mathbf{X}}) := (g(\mathbf{X}^{i_1}), \dots, g(\mathbf{X}^{i_k}))$.

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Desirable properties for an aggregation map for CD:

Property (Aggregation faithfulness)

If $\mathbf{Z}^i \perp\!\!\!\perp \mathbf{Z}^j \mid g(S_{\mathbf{X}}) \Rightarrow \mathbf{X}^i \perp\!\!\!\perp \mathbf{X}^j \mid S_{\mathbf{X}}$, then the map $g(\cdot)$ is said to satisfy the aggregation faithfulness property.

In words, no independence among aggregate variables is accidental.

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Property (Aggregation sufficiency)

If $\mathbf{X}^i \perp\!\!\!\perp \mathbf{X}^j \mid \mathcal{S}_{\mathbf{X}} \Rightarrow \mathbf{Z}^i \perp\!\!\!\perp \mathbf{Z}^j \mid g(\mathcal{S}_{\mathbf{X}})$, then the map $g(\cdot)$ is said to satisfy the aggregation sufficiency property.

In words, aggregation sufficiently captures all independences among $\mathbf{X}^i$'s.

Valid aggregation map definition

Definition (Independence Model)

An independence model $\mathcal{I}(\mathbf{X})$ over a set of variables $\mathbf{X} = \{\mathbf{X}^1, \dots, \mathbf{X}^N\}$ is defined as a collection of a triple of sets of indices $\mathbf{i}, \mathbf{j}, \mathbf{k} \subset [N]$ such that, $(\mathbf{i}, \mathbf{j}, \mathbf{k}) \in \mathcal{I}(\mathbf{X})$ if and only if $\mathbf{X}^{\mathbf{i}} \perp\!\!\!\perp \mathbf{X}^{\mathbf{j}} | \mathbf{X}^{\mathbf{k}}$, where $\mathbf{X}^{\mathbf{i}}$ (resp. $\mathbf{X}^{\mathbf{j}}, \mathbf{X}^{\mathbf{k}}$) refers to the set of variables in $\mathbf{X}$ corresponding to the index set $\mathbf{i}$ (resp. $\mathbf{j}, \mathbf{k}$).

Remark: If $\mathcal{I}(\mathbf{X}) \equiv \mathcal{I}(\mathbf{Y})$, then constraint-based CD yields the same output graph for $\mathbf{X}$ and $\mathbf{Y}$.

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Definition (Valid aggregation map)

A map $g(\cdot)$ over $\mathbf{X}$ is defined as a valid aggregation map if it:

- (i) fulfills aggregation faithfulness and sufficiency or equivalently if,
- (ii) $\mathcal{I}(g(\mathbf{X})) \equiv \mathcal{I}(\mathbf{X})$, where $\mathcal{I}(\mathbf{X})$ denotes the independence model of the set $\mathbf{X}$

Aggregation consistency scores and theorems

- Aim: Construct scores to quantify degree of violation of aggregation faithfulness and sufficiency

Definition (Independence score for aggregation consistency)

The **independence score for aggregation consistency** $c_{ind}(\mathbf{Z}, \mathbf{X}, \mathcal{A})$ given aggregate variables $\mathbf{Z}$, vector-valued variables $\mathbf{X}$, and a CD algorithm $\mathcal{A}$ is defined as follows:

$$c_{ind}(\mathbf{Z}, \mathbf{X}, \mathcal{A}) = \frac{|\mathfrak{C}^{ind}|}{|\mathfrak{C}^{ind}| + |\mathfrak{I}^{ind}|}, \quad (2)$$

where $\mathfrak{C}^{ind} := \{I | I \in \mathcal{I}_{\mathcal{A}}^{test}(\mathbf{Z}) \cap \widehat{\mathcal{I}}(\mathbf{X})\}$ denotes the set of consistent independence statements over $\mathbf{Z}$ that can be verified over $\mathbf{X}$ and $\mathfrak{I}^{ind} := \{I | I \in \mathcal{I}_{\mathcal{A}}^{test}(\mathbf{Z}) \setminus \widehat{\mathcal{I}}(\mathbf{X})\}$ denotes the set of inconsistent independence statements.

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Lemma (Independence score and aggregation faithfulness)

Let CD assumptions and **aggregation graph existence** be satisfied and let CD algorithm $\mathcal{A}$ be **causal input complete**, then the independence score c_{ind} is maximal (i.e. it equals 1) if and only if $g(\cdot)$ satisfies aggregation faithfulness.

Experimental performance of independence score

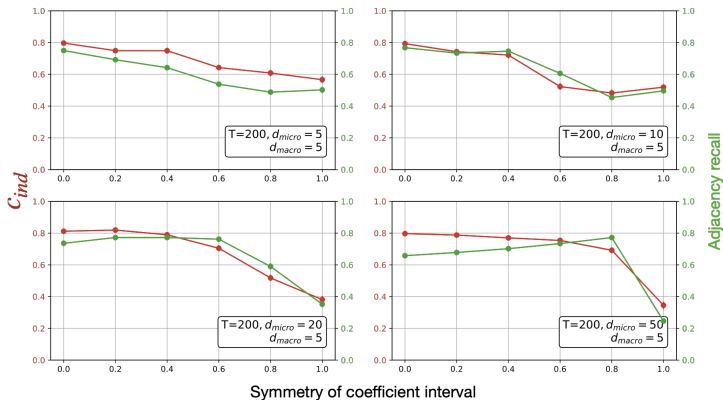


Figure: Independence score c_{ind} captures aggregation faithfulness violations.

Adag Wrapper for vector-valued CD

Central idea behind Adag ('adaptive aggregation'): Tune the aggregation map until you reach a desired aggregation consistency score

- Given a tuning parameter, we define a **tunable aggregation map**, and denote it by $g^{\mathbf{m}}(\cdot) := (g_1^{m_1}, \dots, g_N^{m_N})$ where $\mathbf{m} = (m_1, \dots, m_N) \in \mathbb{N}^N$ is the tuning parameter, and m_i denotes the dimension of the aggregate variable $\mathbf{Z}^i$.
- Eg., if the aggregation map is such that it selects the first n principal components of each $\mathbf{X}^i \in \mathbf{X}$, such that for each i , $|\mathbf{Z}^i| = n$, then increasing n makes aggregate variables higher-dimensional and more expressive.

Experimental Results

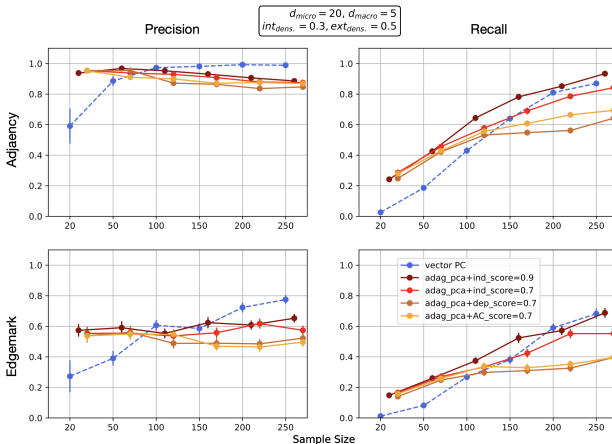


Figure: Adag performance comparison with vector CD: Adjacency (respectively edgemark) precision and recall for vector CD (denoted 'vec') and Adag with independence score c_{ind} , effective dependence score $\bar{c}_{dep}$, and, effective AC score, as sample size increases.

Conclusions and open questions

We looked at:

- Vector-valued CD and pitfalls of typical approaches
- Validity of aggregation for CD
- Checking and optimizing validity using aggregation consistency scores

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We could still:

- Find ways to loosen strict founding assumptions, i.e. SCM existence and faithfulness
- Define aggregation consistency scores more fitting to domain problems
- Construct better multivariate CI tests for specific settings

Thank You!

Contact me at `urmi.ninad@tu-berlin.de`

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